

Central Dogma of Molecular Biology

DNA is transcribed into RNA, which is then translated into proteins

Transcription factor can cause the cell to increase or decrease the production of RNA/protein corresponding to a given gene.

We apply DNA sequencing to various other problems

Let Dna denote a collection of t strings of length n

If we choose a k -mer from each of the strings in Dna then we obtain a collection Motifs

How do we score motifs

We find a consensus string and find differences.

Input: t Dna strings and an integer k w/ a scoring function $Score$

Output: A collection Motifs of k -mers, one from each string in Dna minimizing $Score(Motifs)$

Search space is very big $(n-k+1)^t$

From motifs to a Profile Matrix

(take the frequency of each A, C, T, G per column).

You can score each k -mer with the profile matrix
(multiply each probability)

Keep on picking the best k-mers from the profile

Just because we haven't observed a value doesn't mean its P is 0

We add a pseudocount before normalization

Gibbs Sampling Only one string will be replaced per round

Rather than taking the most probable, we just choose it based on the probability distribution.

Expectation Maximization

Use the Hidden Matrix

To get the profile, you accumulate the Hidden Matrix weights for each profile.

We stop when the profile matrix doesn't change much